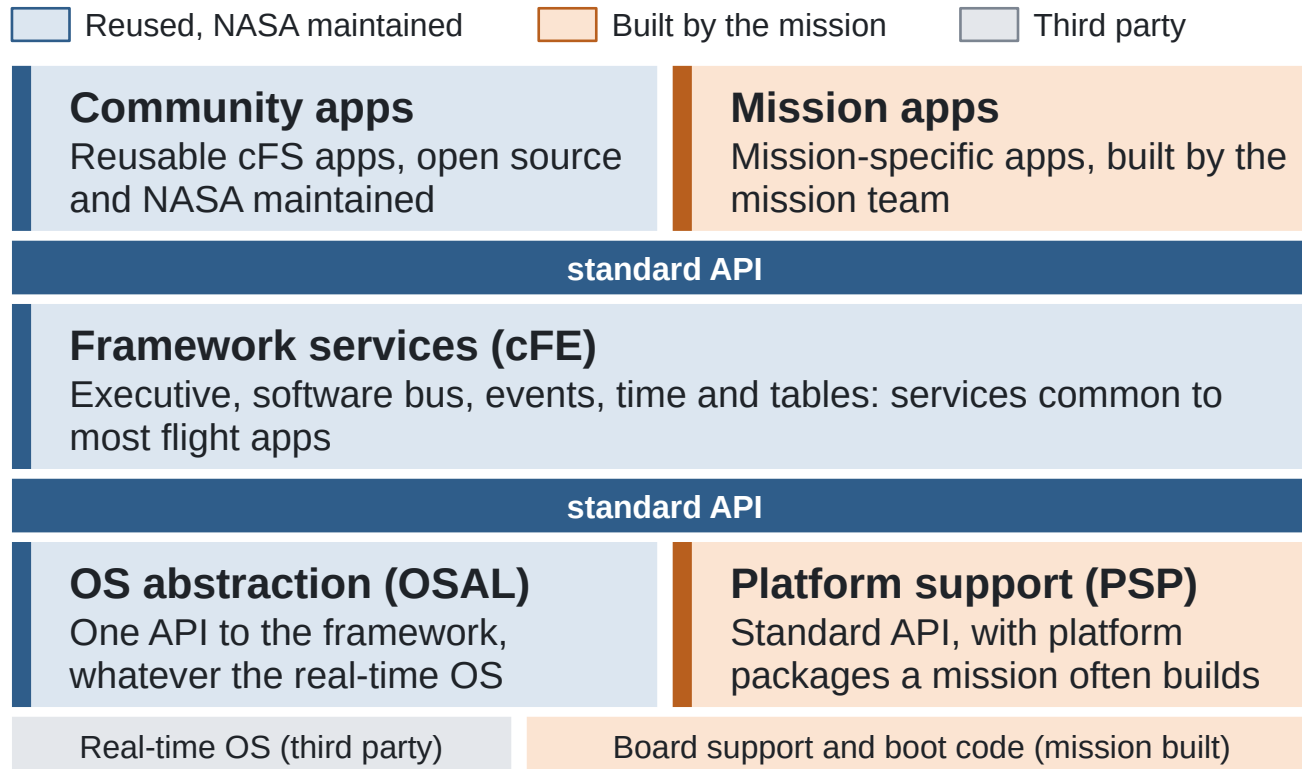


Each layer hides its implementation behind a standard API, so a mission reuses the framework and focuses on custom parts



What NASA says the layers give a mission

Reusable components

"Reuse of tested, certified components supplies savings in each phase of the software development cycle."

Feature: 2015 deck, p8. Quote: 2021 training memo, p41.

Platform independent

"Layered architecture and compile-time configuration parameters make it scalable and portable to a wide range of platforms"

Feature: 2015 deck, p8. Quote: 2015 deck, p2.

New technology without framework changes

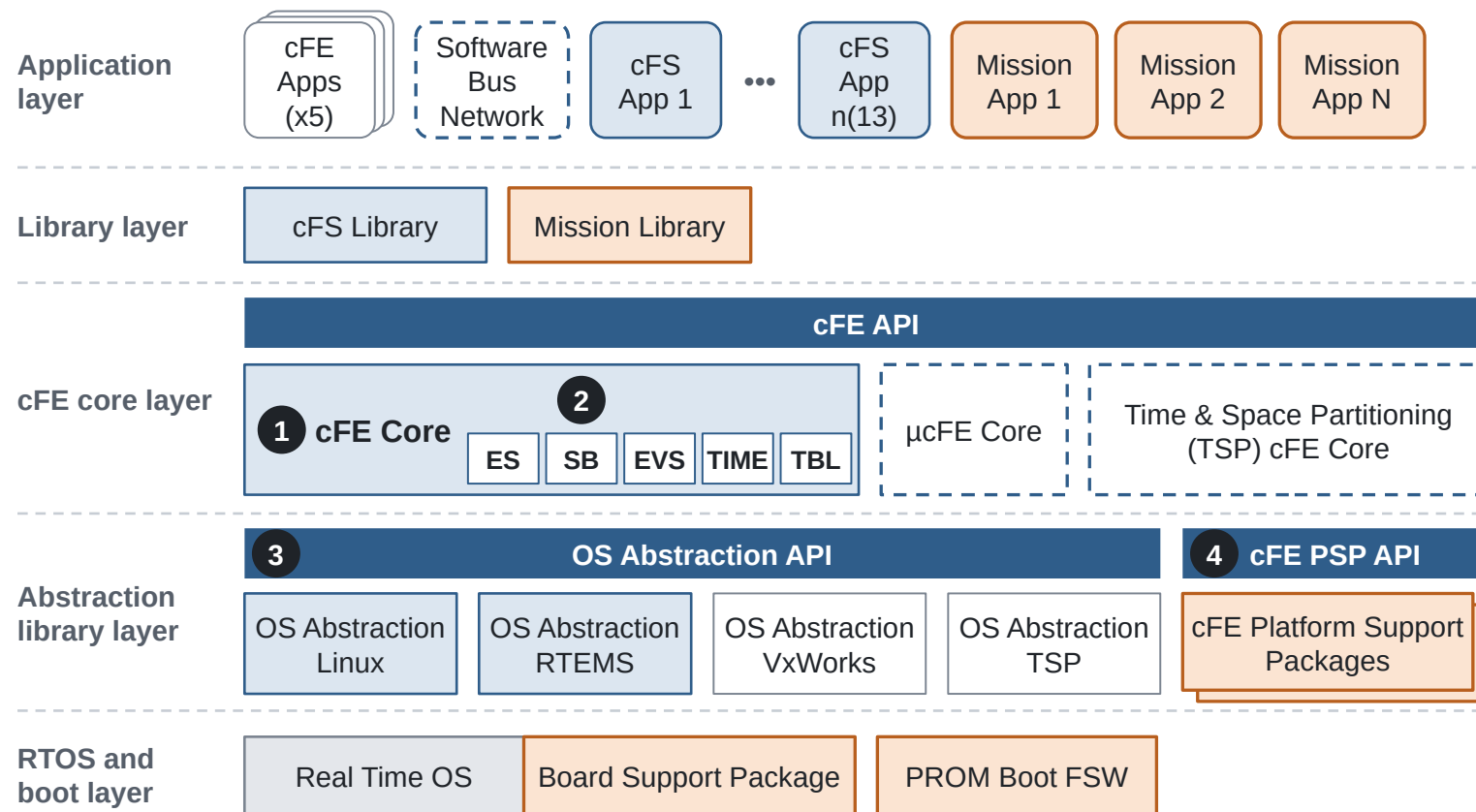
"Internals of a layer can be changed -- without affecting other layers' internals and components."

Feature: 2015 deck, p8. Quote: 2023 overview, slide 8.

Sources: 2015 deck: McComas et al., NASA Goddard, Aug. 2015, NTRS 20150018075. 2023 overview: Baker, NASA Goddard, 2023, NTRS 20230002444. 2021 memo: NASA/TM-20205000691/REV 1, NTRS 20205011588, printed pages. Title: 2023 overview slide 8 and memo p41. Layers: memo pp. 23 to 26 and 45. Colors: 2015 legend (p8) and its note on center PSPs (p23).

Portfolio reconstruction of a public NASA slide. Not a NASA document.

The OSAL and PSP hide the real-time OS and the board, so the cFE and the apps above it stay platform independent



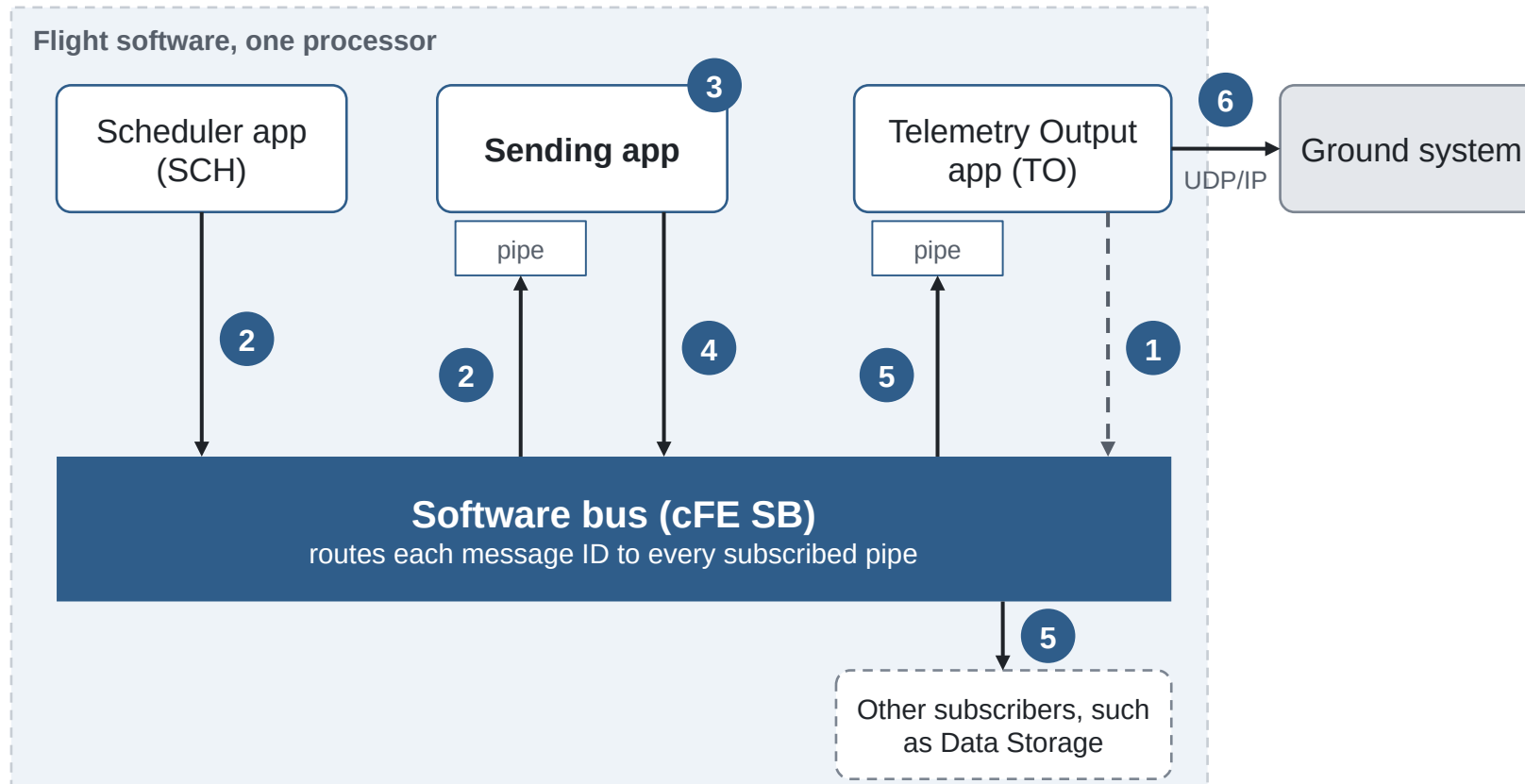
- 1 cFE Core services** pp. 25, 45, paraphrase
Provide the services common to most flight apps: ES, SB, EVS, TIME and TBL
- 2 Software bus (SB)** p45
“Provides an application publish/subscribe messaging service”
- 3 OSAL (OS Abstraction API)** p23, paraphrase
One API to the cFE, whatever real-time OS runs underneath
- 4 PSP (cFE PSP API)** p24, paraphrase
One API to the avionics hardware and board support package

Status as of August 2015

Open source, NASA maintained	Mission developed
In development	Third party
No status color in source	Layer API

Source: diagram rebuilt from McComas et al., NASA Goddard, Aug. 2015, NTRS 20150018075, p8. Service acronyms and order: 2021 memo, printed p45 (the 2015 deck lists the same five services on p9). Roles: NASA/TM-20205000691/REV 1, Jan. 2021, NTRS 20205011588, printed page on each heading. Dark bars: each layer's standard API (2023 overview, NTRS 20230002444, slide 8).

One housekeeping packet crosses the software bus in six steps, and the app that sends it does not know who subscribed



- 1 Subscribe.** At startup, TO subscribes a pipe to the packet's ID.
2021 memo, p85 and p93
- 2 Request.** The scheduler sends a housekeeping request over the bus.
2021 memo, p86 and p34
- 3 Build.** The app fills its housekeeping packet and stamps the time.
2021 memo, p88 and p98
- 4 Send.** The app sends the packet without knowing who subscribes.
2021 memo, p85 and p97
- 5 Route.** The bus routes it to every pipe subscribed to its ID.
2021 memo, p85
- 6 Downlink.** TO reads its pipe and sends the packet over UDP/IP.
2021 memo, p95 and p34

Sources: steps from NASA/TM-20205000691/REV 1 (the 2021 memo), Jan. 2021, NTRS 20205011588, printed pages. TO in the open-source bundle is a lab app meant as an example (nasa/cFS README). The memo's CFE_SB_SendMsg and CFE_SB_RcvMsg (p97) are CFE_SB_TransmitMsg and CFE_SB_ReceiveBuffer in the current cFE Application Developers Guide (nasa/cFE, sec. 6.6 and 6.7, read Oct. 6, 2026).

Portfolio diagram drawn from public NASA documents. Not a NASA document.